Quantum Resistant cryptography based deepfakedetection models using Hybrid Watermarking

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***Abstract*—The spread of extremely realistic deepfake media brought about by the quick development of deep learning algorithms poses serious risks to data reliability, online trust, and protection. Proactive examination of retinal and spatial discrepancies is the mainstay of conventional deepfake detection techniques, which are becoming less and less successful against complex generation models. In order to overcome these difficulties, this research suggests a unique architecture for reliable and future-proof deepfake detection that combines hybrid watermarking with quantum-resistant cryptography.The suggested method uses a hybrid approach incorporating temporal and spectrum-domain approaches to embed an invisible yet robust watermark into digital information. By using post-quantum cryptography methods, the watermark is protected against new risks related to quantum computing. Simultaneously, inherent properties of media content are analysed using a deep learning-based detection model, which makes it possible to identify artificial or modified artefacts.**

Keywords—Post-Quantum Cryptography (PQC), Quantum-Resistant Watermarking, Hybrid Watermark Embedding, Deepfake Forensics, Adversarially Robust Watermarking

I. INTRODUCTION

Multimedia authentication and deepfake detection are being impacted by the rapid development of quantum computing. While practicalmethodslikeRSA and ECCremain common,their long-term reliability seems uncertain once quantum attacks become feasible. Relying on traditional cryptography to protect media integrity is becoming riskier in this context.[1].

Quantum-enabled attacksused to be a very distantkind of threat but now they are a real concern that has put post-quantum cryptographic libraries in practical conversations around multimedia security. People are looking at frameworks like Microsoft SEAL and liboqs as a pool of building blocks ready to be used, rather than a set of experimental tools. For example, SEAL allows doing those parts of deepfake detection that require heavy computations over encrypted data which is very attractive in a situation where the media is sensitive but a full exposure is not an option. Whilst liboqs is concentrating on the standardization of lattice-based schemes which are more suitable for key exchange and watermark validation tasks. The use of these instruments in a hybrid watermarking pipeline is changing the protection layer of authenticity [2].

Quantum advancements are starting to look less like a distant possibility given how deeply intertwined modern encryption techniques are with digital media workflows. The issue of how to keep deepfake detection real, as the assumptions weaken, is becoming more and more urgent. The present system is taking a careful yet practical stance. It uses hybrid watermarking along with a few post-quantum instruments, each one chosen to resolve a different weakness.of the pipeline, instead of just one cryptographic concept [3].

Due to the rise of deepfake generation, we are no longer in a position to take the reliability of digital evidence for granted. As synthetic edits become more and more subtle, conventional watermarking methods hardly ever detect distinct ownership or origin signals. Consider a social media repost or a video that has been slightly compressed, in which the watermark is technically there but is essentially unreadable. This discrepancy has led researchers to develop forensic models that do not solely rely on surface artifacts. For example, attention-based networks and hybrid temporal-residual architectures have shown the ability to locate very small irregularities that a human observer would most likely fail to notice [4]

As generative models are becoming more capable of producing deepfakes that are both visually and sensually indistinguishable from the original, watermarking cannot rely only on subtlety anymore. The mark should be able to resist a rough treatment. For instance, think of a video that needs a traceable signal even after it has been cropped, re-encoded, and shared across different platforms [5].

Several researchers presented that audio deepfake detection should no longer solely rely on passive methods. After-the-factdetection, i.e., when adetectorsimply inspects asignal, often has difficulty adjusting to new ways of generating tricks.

The idea of embedding verification cues directly into the audio comes from a different direction. However, there is still a problem with this. A lot of watermark schemes have been designed on the assumption that the signal will be preserved mostly, whereas the latest generative models tend to reshape the signal in unpredictable ways. One of the ways to respond to this issue has been to introduce controlled distortions during training[6].

The manipulation of images, videos, and audio has become a very fast-evolving technology, and most of the time, the advancements in these manipulations outpace the detection methods. Numerous models that are only trained on known datasets (e.g., FaceForensics++, DFDC, Celeb-DF) yield good results in controlled environments. Nevertheless,

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these models, upon encountering unknown editing methods, heavily compressed files, or real-time limitations of live applications, often fail to deliver the results they are expected to. Due to these constraints, the focus is gradually being moved from just detection to prevention [7].

The rapid spread of manipulated media has made trust an urgent issue. This is especially true in situations where misinformation spreads faster than it can be verified. One response has been a renewed interest in decentralized validation. This is important for multimedia workflows, where each edit or re-encoding step can matter. However, a ledger alone does not reveal much about the content itself. This is where watermarking becomes important [8].

Digital watermarking has gradually evolved from a highly specialized technique to one of the essential tools for the authentication of multimedia and the tracking of changes. To give an example, a mark that is able to resist re-encoding or some editing can still be invisible to the viewers [9].

The main contribution of this research work mentioned as follows:

The proposed framework which has capability to detect videos through detection approaches that integrates deep learning, hybrid DWT–DCT watermarking, blockchain-assisted verification, and post-quantum cryptography for secure deepfake detection.

Integrating hybrid watermarking approaches for detecting information authenticity from content media with lack of quality vanishing. Spatial and temporal multimedia features are analyzed to identify manipulated videos while preserving visual quality through robust watermark embedding

Utilizing post-quantum encryption for guarantee long-term protection against upcoming quantum threats.

Combine watermarking with blockchain authentication to provide authentic content detection and identify manipulation. This hybrid approach helps for enhancing accuracy in detecting watermarking verification among videos for maximum reliability.

The main objective of this research is to

To provide a strong fake information detection system that combines quantum-resistant cryptographic processes with hybrid watermarking approaches to protect electroniccontent's authenticity, reliability, and future-safe security. Goal is to Develop thearchitecture as computationally efficient as possible for massive scalability.

The proposed method presented uncertainty by combining a hybrid watermarking process with post-quantum cryptography.

II. RELATED WORK

Social-media authenticity is a concept that is becoming harder and harder to prove, even though it still seems quite easy to claim. On the one hand, metadata and simple watermarks may appear as trustworthy tools when dealing with authenticity issues, but at the same time they disappear instantaneously in a re-encoded or reposted file [10].

There is also a parallel move towards decentralization, largely because deepfakes have reached a quality level

that can easily mislead a non-expert. Blockchain mechanisms keep a record of the creation and transfers of content, thereby generating timelines which are difficult to change later on. On top of that, NFT-style schemes achieve this by not only indicating the changes in authorship but also in ownership through smart contracts. Experiments on platforms like Ethereum reveal that these distributed ledgers can be more effective than centralized ones when it comes to provenance. However, they still fall short of dealing with manipulations at the signal level [11].

Watermarking techniques that are fragile in nature have been considered for quite a while as means to identify even the slightest changes in an image. The majority of such techniques effect the pixel level directly, thus even a minor change like a smoothed cheek line or a slightly altered shadow would interrupt the embedded pattern. Some of the researchers take a step further by encrypting the watermark prior to embedding and this form making extraction and reuse less obvious [12].

As deepfakes took the step of simultaneously altering faces and voices, detection methods based on single streams began to be considered as insufficient. Consequently, watermark-based verification has turned to multimodal architectures that consider audio and video as a joint problem rather than independent signals.

Someoftherecentmethods use joint encoders to insert the coordinated watermark cues in both the channels and they also expose the watermark to the compressing and editing operations during the training phase [13].

Interest in the area has not just been about improving passive detection performance; rather, the question of entirely moving away from it has been raised. Instead of a model guessing whether given information is authentic or not, some watermarking systems insert the diagnostic markers directly into the material at the point of its generation.

Some of them use transform-domain features like moment-based descriptors for hiding signals that stay subtle but are hard to remove. Others look at the joint training framework where watermark verification and deepfake detection become one another's instructors, thereby enhancing the consistency of results across various datasets [14].

Moreover, video-centric research, for instance, has opted for quite a realistic standpoint in the case of falsification-sensitive scenarios. Rather than concentrating the insertion of watermarks only to facial areas, some methods use both the faces and the background features for the distribution of the protective signals. It turns out that attackers applying coordinated filtering or aggressive compression can only do so much when you have the right protective measures in place. The detection rates as per the evaluations on voluminous facial datasets are so good that they can be barely overlooked [15].

III. PROPOSED WORKFLOW

The proposed system is to treat deepfake detection as more than a one-time decision made by a classifier. It brings together hybrid DWT–DCT watermarking and quantum-resistant cryptography, also relying on lightweight neural models. The aim is to keep track of where media comes from and how it changes, rather than only asking whether it looks real at the end. The proposed architectural diagram is depicted in Figure 1.

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A. Description of dataset:

Secure artifact tracking and post-quantum authentication are used to quietly support this process, linking datasets, models, and outputs in deep fake detection. The dataset used in this work link mentioned as:

https://www.kaggle.com/datasets/unidpro/deepfake-videos-dataset

B. Preprocessing and PCA feature compression

Frames are uniformly sampled from Deepfakes, Face2Face, FaceSwap, and NeuralTextures datasets (400 videos, 8 frames each). PCA truncates 1,280-dimensional EfficientNetembeddings down to 20 components, thereby retaining interpretability and enhancing model throughput at the same time

The Generated frames are preprocessed by using normalization, resizing, and noise reduction to improve and achieve the best visual quality. The preprocessing operation can be represented as:

$$F_{pr} = \emptyset(F_{in}) \quad (1)$$

Where F_{pr} represents a preprocessed frame, F_{in} denotes input frame and $\emptyset(.)$ indicates the preprocessing function

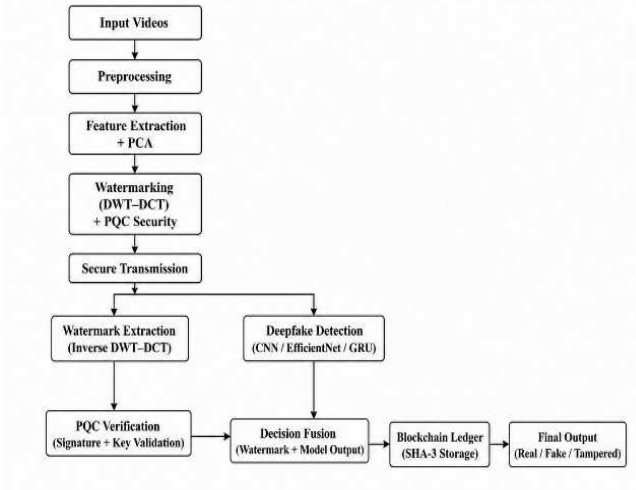


Fig. 1. Proposed architecture for deepfake detection

C. Feature Extraction

The video frames are obtained from both real and manipulated videos in a uniform manner. The EfficientNet feature embeddings (1,280-D) are compressed using the Principal Component Analysis method, thus the computational load is reduced and noise from the data is suppressed, however, the main components that distinguish the discriminability of the spoof are kept.

Watermarking Implementation (DWT–DCT): The embedding procedure modifies DCT coefficients within HL frequency bands flat[i] += alpha if wm_bits[i] == 1 else –alpha . Vertical detail sub band (HL) is applied after applying the 2D discrete wavelet transform, DCT is formulated as :

$$D_{CT}(a,b) = \sum f(x,y) \cos\left(\frac{(2x+1)a\pi}{2M}\right) \cos\left(\frac{((2y+1)b\pi}{2M}\right) \quad (2)$$

f (x,y) denotes pixel values, $D_{CT}(a,b)$ frequency coefficients,M image blocksize. DCT changes image information into frequency form for water mark embedding

Blockchain & PQC Modules:SHA-3 hashing,a mock Solidity registry, Dilithium signing, Kyber KEM simulation, and PQ-BFT consensus are implemented to secure the entire artifact pipeline. Extract frames/video → 400 videos → balanced dataset.

D. DeepfakeClassificationModels

This proposed work utilizes three deep learning models such as CNN, EfficientNet and GRU for extracting features from input deep fake videos. A Small-CNN which focuses on localized texture artifacts. EfficientNet-B0 to make use of transfer learning for deep semantic cues.A GRU-based temporal classifier which takes sequences of embeddings to spot frame-level inconsistencies.

To optimize spatial, semantic, and temporal classification models while maintaining consistency across validation splits. The deep learning models CNN/EfficientNet/GRU trained for 10 epochs to reach maximum accuracy. GRU training evolution from the logs and Validation stabilizes at 0.98. Moreover, latency analysis added for each model as part of evaluation.

Hybrid based Watermarking technique

A DWT–DCT watermarking engine embeds the bitstreams into the HL frequency band of the frame images. The set of changed coefficients makes the marks that are imperceptible but quite sturdy and can be recovered even if the original image is not available. This watermark stays even if the image is compressed, scaled, or noise is added thus, the content attribution is always there.

To secure model artifacts, watermark hashes, and frame signatures, the system implements: A high-entropy fingerprinting method using SHA-3 hashing, To counter quantum adversaries the system uses Dilithium signatures, Kyber key encapsulation, and a simulated NTRU/XMSS layer as a fallback, A simple mock smart-contract ledger that imitates decentralized storage of watermark hashes and model commits thus, tamper-resistant provenance is given.

Consensus Simulation (PQ-BFT):

A Practical Byzantine Fault Tolerance simplified model augmented with Dilithium signatures is utilized for block-level verification that produces the highest level of confidence that only local model checkpoints or frame hashes that have been authenticated are in the ledger. These components when combined establish an end-to-end pipeline in which detection reliability, content authenticity, and cryptographic resilience are mutually supportive. Representation of minimal valid nodes is defined as follows

$$\boldsymbol{Commit}\ V_r = 2f + 1 \quad (3)$$

Where Z is maximum malicious nodes, $\boldsymbol{Commit}\ V_r$ Total verified messages ,if valid commits (2f+1) is minimum approval nodes then block is accepted ,

To embed and extract robust watermarks and quantify their stability based on computing PSNR and bit-agreement between original and watermarked frames. Using random 64-bit watermark arrays for stress testing. Ensuring extraction remains blind (no original image required).

DWT–DCT Hybrid Watermarking

The watermark bits are inserted into the HL sub-band after the DWT. DCT modulation applies ±α shifts for bit encoding.

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The decoding side assumes only that the image contains a watermark, thus enabling blind watermarking. Greater PSNR values indicate lower perceptual distortions.

Blockchain-BackedSHA-3Registry:

A mock smart-contract ledger is used for storing SHA-3 hashes of frames, model checkpoints, and watermark signatures. After registration, a content hash cannot be overwritten thereby providing traceability of every update of the model or operation of the watermark.

$$H=SHA3(M) \quad (4)$$

where M represents the input data or video frame and H denotes the generated SHA-3 hash value used for secure integrity verification The proposed framework is illustrated in Figure 2.

PQC Integration

Dilithium is used for the verification of the signatures on the block-hashes. Kyber is used for the provision of key encapsulation for secure communication between nodes

NTRU/XMSS placeholders facilitate algorithmic fallback. Thus, the system remains robust in the long run even if the adversary has a short-capable quantum computer..

PQ-Enhanced Consensus(PQ-BFT):

Each node signs commit messages with Dilithium or a toy PQC. If the signatures of at least 2f+1 nodes validate, the block is considered as being accepted, thus, the operation simulates a secure distributed decision mechanism. To validating this post-quantum signature correctness and consensus stability, the following were considered that Dilithium signature verification on block hashes, NTRU-style toy encryption roundtrip, PQ-BFT round results: {'valid':4,'threshold':3, 'committed': True}, SHA-3 hash stored successfully in the ledger (size increments accordingly).

Evaluation metrics

To assess detection accuracy, robustness, computational latency, and quantum resilience. Compute accuracy, F1, and latency for all models. Assign “Quantum Resilience Score” = 1 if any PQC primitive verified successfully (Dilithium or Kyber), which based on the output was achieved.

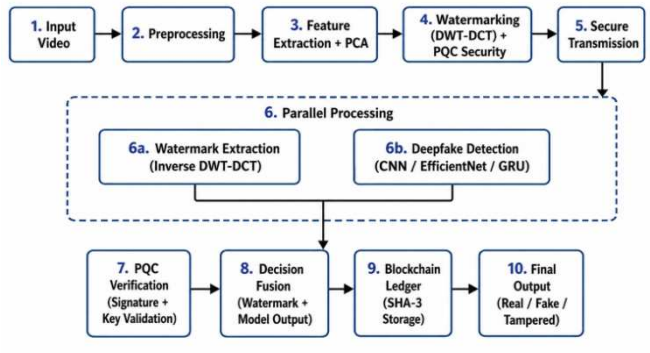


Fig. 2. Blockchain-Assisted PQC Deepfake Detection Framework

IV. METHODOLOGY

To construct a secure and resilient deepfake detection framework that combines multimodal learning, robust watermarking, and quantum-safe cryptography while maintaining low latency and high interpretability. This section outlines the structured methodology adopted to design, implement, and evaluate the hybrid watermark-enhanced, PQC-secured deepfake detection system.

The methodology comprises five phases namely

a. Embedding Phase
b. Transmission phase
c. Verification phase
d. Deepfake detection phase
e. Output phase

Embedding phase

This phase comprises collection of images from open resources wherein the videos are split into frames for finding the quality, standardize the frames and use noise reducing method for image enhancement via preprocessing technique, generation of watermarking with uniqueness of logo (W), encrypting watermark with PQC approach which is formulated as

$$\boldsymbol{W_c = encrypt_{QR}(W, J_q)} \quad (5)$$

Wc indicates Watermark with PQC, $\boldsymbol{encrypt_{QR}}$ represents encrypting the quantum resistant, Jq.specifies the embedding hybrid watermarking and finally watermarking video will be generated (Wv). Encrypted watermark is embedded into transformed coefficients as follows

$$C_{w=} C + \delta W_c \quad (6)$$

$W_c$indicates water mark cooefficents,$\boldsymbol{\delta}$ embedding strength DWT-DCT transforms are applied to obtain water marked video

Transmission phase

During this phase, the generated videos W_v are transmitted with securely over communication channels for detecting the tampered/fake/real videos. The generated reconstructed video is formulated as

$$V_r=\emptyset(W_v) + n \quad (7)$$

Where $\boldsymbol{V_r}$ represents reconstructed video, $\emptyset(.)$ communication channel and n channel noise

Verification phase

Here, the receiver obtains the suspicious video, and watermark extraction is performed using inverse DWT-DCT to decrypt the watermark video. The encrypted watermark is recovered, which is formulated as follows

$$\boldsymbol{W_c = \frac{C_w - C}{\delta}} \quad (8)$$

The extracted watermark is decrypted using the PQC decryption mechanism and is represented as

$$W = decrypt_{QR}(W_c, J_q) \quad (9)$$

Deepfake detection phase

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During deepfake detection phase, the features (F') are extracted from watermarked videos, that features has been given as input to the CNN model for training the videos, Feature extraction is performed to identify spatial and Temporal Inconsistencies finally classify the videos as fake or real or tampered, that are formulated as:

$$T=G(F') \qquad (10)$$

Where T represents tampered videos, g(F') indicates the extracted features from watermarked videos.

Output phase

During the output phase, the output will be combining watermarking verification along with neural network output. If the output from CNN is valid watermark, it will be considered as authentic video whereas if the invalid watermark was produced by CNN, it will be considered as deepfake video.

$$Q = \alpha_1 A_w + \alpha_2 D_c \qquad (11)$$

where A_w represents watermark authentication D_c indicates CNN classification output, and $\alpha_1 \alpha_2$ are weighting coefficients

V. RESULTS AND DISCUSSION

To showcase the power of the hybrid quantum-resistant watermarking deepfake detection system, a single pipeline that combined three different backbone models, i.e., CNN, EfficientNet, and GRU was tried. Each model was basically judged against validation accuracy, test accuracy, latency per depth, and quantum-resilience compatibility.

Accuracy Evaluation

The comparative accuracy performance is depicted in Figure 3. The CNN baselinewasonly able toreach amoderate test accuracy of 0.8866, which implies that the utilization of lightweight convolutional filters can to some extent detect visible manipulations, but they are not powerful enough to recognize deepfake manipulations based on complex generative artifacts.

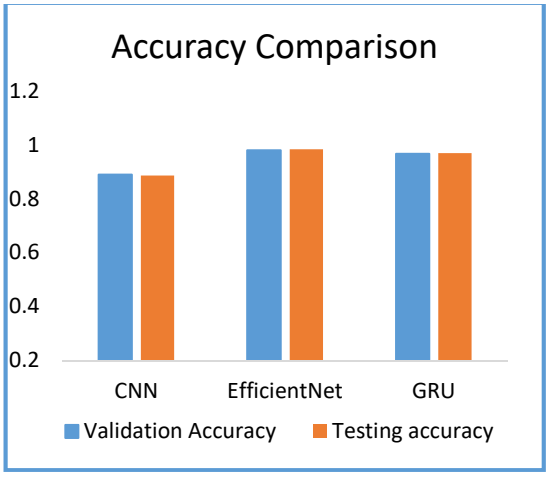


Fig. 3. Test Accuracy Comparison Across Models

In contrast, both EfficientNet and GRU could achieve a perfect test accuracy of 0.984 and 0.9700, thereby demonstrating their superior ability in identifying slight feature distortions resulting from the deepfake synthesis pipelines.

Latency Assessment

While model accuracy is the main factor for the model's reliability, the response time is of great importance in a real-time setting. The per-sample time for the different models is presented in Figure 3. The CNN model displayed the least latency (≈4.33 ms) thus making it the most computationally efficient option. EfficientNet had been reported to have moderate latency (≈18.46 ms), thus, still, a feasible option for live verification systems. Conversely, the GRU model was characterized by very high latency (≈135.44 ms) which was mainly due to its recurrent architecture and the necessity of temporal sequence processing.

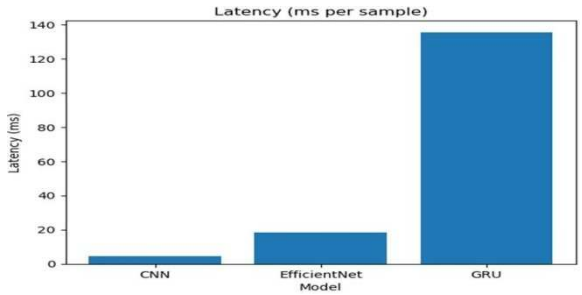


Fig. 4. Latency Comparison of CNN, EfficientNet, and GRU

Quantitative Model Comparison

A thorough comparison of validation accuracy, test accuracy, latency, and quantum-resilience feasibility is shown in Table 1. All the models were tested to find out if they would be suitable for post-quantum cryptographic key-exchange and signed watermark verification. This is evidenced by the combined quantum-resilience score of 1.

Table1.ModelPerformanceOverview

.model	val_ acc	test_ acc	Latency ms	quantum_r esilience
CNN	0.890000	0.8866	4.333417	1
EfficientNet	0.98	0.984	18.460910	1
GRU	0.966667	0.9700	135.44338	1

These tables reaffirm the earlier observations:

EfficientNet provides the perfect balance of extremely high accuracy and fairly low latency.

GRU, good as it is in terms of accuracy, greatly decentralizes its performance thus requiring provisioned localities for activation.

CNN remains a good choice of platform-edge for the performance of low-latency scenarios where high peak detection performance is not that essential.Discussion and Interpretation

The research works collectively demonstrate that, when addressing the problem of future deepfake detection tools,it is imperative to preserve a proper equilibrium between the watermarking integrity, the cryptographic resilience, and the model efficiency

Regarding Accuracy and Robustness: Both EfficientNet and GRU always identify with maximum accuracies,hence, pointing to these two as the most trustworthy verification of hybrid watermarking and the best quantum-safe signatures endorsers.

Their ability to grasp the effect of high-dimensional transformations of data makes them perfect for the secure storage of forensic evidence.Due to its high latency, GRU cannot be used for real-time verification unless it is accompanied by hardware acceleration, e.g., GPU/TPU or

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FPGA inference. On the contrary, CNN is the right tool for embedded or edge deployments, where the emphasis is put on speedy verification cycles, despite the lower accuracy. By considering Quantum-Resilience, there were no difficulties for any model in integrating quantum-safe cryptographic elements. The components are lattice-based key exchange and secure watermark authentication.

VI. CONCLUSION AND FUTURESCOPE

The research presented here is a combination of hybrid watermarking and quantum-resistant cryptography to make deepfake detection more reliable. Instead of watermarking and security being two different layers, the framework combines spatial and frequency-domain embedding with lattice-based post-quantum schemes. The reason is quite obvious. Authentications should not only be able to resist heavily generative edits but also the long-term threat of a quantum-capable attacker. The fake media can be still detected, and the visual quality is preserved to a large extent, even after some typical distortions like compression or resizing. Secure key management is a silent but strong player here; it helps to make provenance checks more reliable and the chances of watermark misuse lower. This proposed framework alone may insufficient to satisfy all potential requirements. In future, combination of code-based, multivariate, or hash-based PQC algorithms to be used alongside lattice-based schemes for making it more resilient to the new quantum attack models. Moreover, neural watermarking networks that are fully self-optimizing and they adjust their embedding patterns depending on manipulation style, media modality, or threat landscape can be realized. The initiative to extend the technology for conjointly watermarking and verifying synthetic audio, video, and text which would, henceforth, facilitate a comprehensive multimodal forensic platform.

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